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Ballincollig Royal Gunpowder Mills: the vaulted magazine

By GILLIAN BOAZMAN

ABSTRACT: A consideration of the historical context, form and function of the vaulted magazine at Ballincollig Royal Gunpowder Mills.

Ballincollig Royal Gunpowder Mills manufacturing complex covers an area of over 400 acres south of the River Lee at Ballincollig, which is situated 5km west of Cork City. The vaulted magazine is one of two magazines in the complex.

HISTORY OF THE SITE

The international and national political situation in the early 1790s prompted Charles Henry Leslie to see an economic opportunity in building a gunpowder mills at Ballincollig. The French revolutionary government had declared war on England in February 1793. Despite the Board of Ordnance having acquired the Home

works at Faversham Kent in 1760 and Waltham in 1788:

Private manufacturers were always needed in wartime and the Board of Ordnance never lost its function as a purchasing rather than a manufacturing agency. (Coad 1989, 245)

The capacity for manufacturing gunpowder in Ireland was minimal: two small mills in the Clondalkin area, which produced scarcely enough for quarrying and sporting needs within Ireland. The United Irishmen were founded by Tone in 1791 and this unstable internal political situation caused anxiety on the part of the

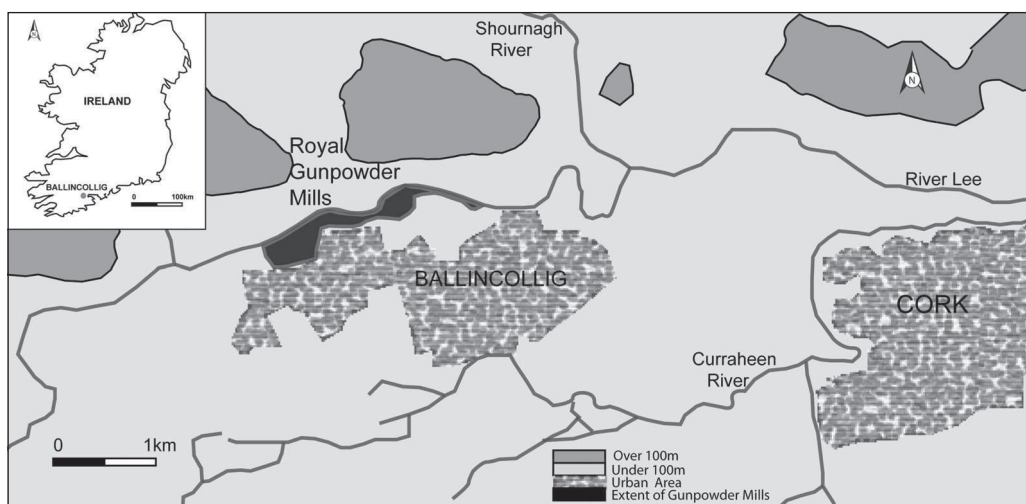


Fig. 1. From OS Discovery Series map 1:50000: position of site

English government about the clandestine import and insecure storage of gunpowder. A proclamation of 1792 banning movement of gunpowder led to the Gunpowder Act of 1793 which prohibited import and transport of all types of weapons and gunpowder.

Leslie thus had a market. He also had land next to a source of waterpower: the River Lee and its overflow channels. He had capital: his father was a doctor and banker, who died in 1793. Finally he had a source of cheap, skilled labour, Kelleher (1993, 21) refers to stagnation of trade in Cork in 1793 and the fears of 'commotion among the manufacturing poor'. The site also had the advantage of being in a sparsely populated area, for in the history of gunpowder manufacture there is constant reference to the danger to the surrounding population. For example, long before Leslie's Cork venture, in the early 18th century, public anxiety caused the movement of Ordnance storage from the Square Tower in the city of Portsmouth, to Priddy's Hard at Gosport (Coad 1989, 260).

Leslie utilised an existing channel for a canal and built two incorporating mills, processing houses and mills for basic materials, pressing

and drying houses, a magazine and ancillary trade buildings: cooperage, millwright's workshop and a forge (Rynne 1999, 148, Fig. 73).

Against a background of declining international relations in Europe, the prospect of war with Napoleonic France encouraged the Board of Ordnance to invest further in manufacturing capacity, and as part of this policy bought the Ballincollig complex, in 1805, from Leslie. Charles Wilks was appointed superintendent on 9th March 1805 (Kelleher 1993, 32). Wilks had previous experience at Chatham and as clerk of works at Faversham and Waltham, and he undertook a substantial expansion of the powder mills in the first year of Board of Ordnance ownership, as can be seen from an inventory of present buildings signed by Wilks in 1806 (WO 55/2285). He added four incorporating mills in 1805 and a further two in 1809. Subsequently Wilks extended the primary materials buildings in the area utilised by Leslie. He initiated changes in technology, for example charcoal was no longer burnt in pits but in cylinders called retorts. Increased energy to power these developments came from a mill race from the Lee just east of Iniscarra Bridge.

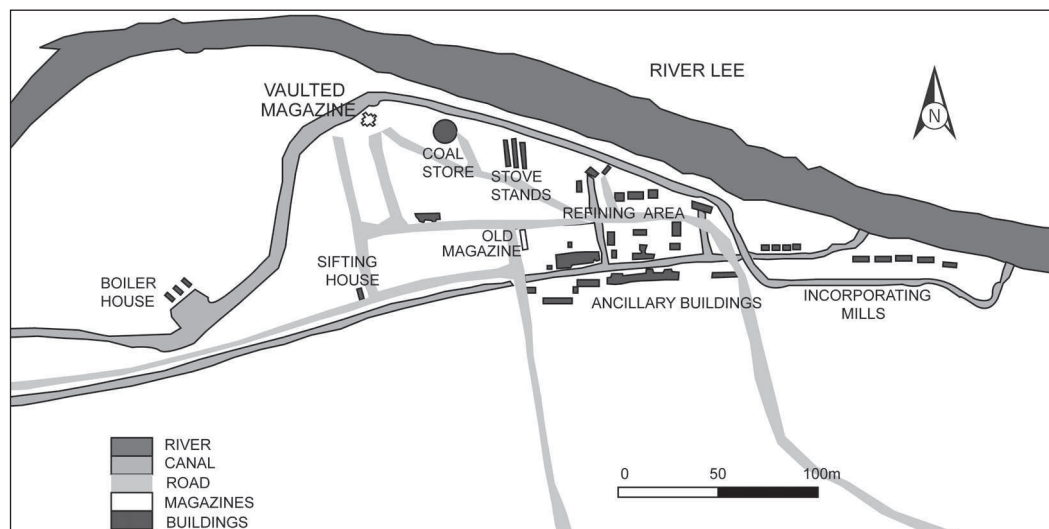


Fig. 2. Redrawing of 1st edition map indicating buildings extant in 1834 when the Gunpowder Mills were purchased by the Tobins

This mill race ran the whole length of the site to the incorporating mills at the eastern end. Due to the volatile nature of the ingredients the finishing buildings were spread out over the two kilometres to the west. Water was utilised again in an extensive canal system to move the product safely around the site. After the battle of Waterloo in 1815, demand for powder dropped dramatically and the Ballincollig mills fell into a state of disrepair: machinery from the incorporating mills was removed to Waltham, causing considerable damage to the mills. In 1828 the Board of Ordnance commissioned a survey of their property, obviously with a view to finding a purchaser.

A group of Liverpool merchants including John and Thomas Tobin and Thomas Tobin Junior had been involved in slave trading and latterly the importing of palm oil. The Abolition of Slavery Act in 1833 meant that they were searching for a new economic enterprise and in 1834 after a lengthy negotiation they purchased Ballincollig gunpowder mills from the Board of Ordnance. The economic *raison d'être* behind this move was that they could continue to optimise their African network, exporting powder and cheap guns and importing palm oil.

When the Tobins purchased the mills (Fig. 2) they were in a poor state of repair. The canals had to be cleaned and there were two incorporating mills added in the 1840s and two more in the 1850s. They also built a larger sawmill powered by a turbine and a much larger drying house. The third magazine and wooden trackway were built at this period. Although the African connection accounted for the greater part of the production, railways and quarrying in Ireland were other outlets. The period from 1850 to 1870 is marked by concern of various parties about the risk that Ballincollig gunpowder mills posed to first the general public as the gunpowder was transported through the streets of Cork to the quays and secondly to Ballincollig garrison from the overstocked magazines. The first problem was

never resolved as Thomas Tobin was an adroit politician and to the end continued to ship his powder from the quay most convenient to him. The second problem was resolved by an inspection team from The War Office in 1870 who suggested the removal of the masonry roof of the magazine built by Wilks (Kelleher 1993, 63-71).

Developments in explosives technology from the 1850s meant that black powder was superseded by gun cotton, nitro-glycerine and cordite. The African connection was not using as much powder and finally in 1888 the gunpowder mills were sold to a consortium of business men. Despite attempts to make specialised powder for Martini-Henry rifles, production went down and the company was taken over by Curtis's and Harvey. The mills were closed in 1903.

THE CHRONOLOGY OF MAGAZINES AT THE ROYAL GUNPOWDER MILLS

The process of gunpowder manufacture is well described by Rynne (2006, 292-294) and Hamond and Hegarty (1987). Thus this article will focus on the last stage of the process which is storage of the finished product. Gunpowder was stored in white oak barrels, approximately 45kg weight, which were manufactured and maintained at a cooperage on site. The barrels were then transported by canal to the magazine to await dispatch. The derivation of magazine is the French *magasin*, a store (Oxford English Dictionary 2005). The two main attributes of a magazine was that it should be both dark and dry, thus the magazines at Ballincollig were windowless and had an underfloor ventilation system. There were two lines of thought on their fabric, either thick walled to absorb explosion or lightly built to minimize damage from flying debris if ignition took place. This issue is discussed below.

The first magazine was one of Leslie's original buildings. In the 1806 survey of the gunpowder mills signed by Charles Wilks, the dimensions of the magazine are 57' x 18' 6" and

it is described as 'stone and slated' (WO 55/2285, 'o'). It lay to the east of the saltpetre refinery. It is designated in the 1828 inventory as 'C' 'Old Magazine' and described as being 'in tolerably good external and internal repair'. It is hard to make out on the 1876 map and is named as a cart shed on the 1891 map. These are official Explosives' Inspector's maps: it would no doubt be listed as a magazine if it was functioning as one. There is no trace of the building now, except a low hump about 12m long which was possibly the western wall of this original magazine.

The vaulted magazine, the subject of this article, does not appear on Rynne's map of Leslie's original manufacturing buildings (1999, Fig. 73, 148). However it does appear

on the 1806 Statement of the Property of the Ordnance at Ballincollig (WO 55/2285, No 8) which was a report to the Board, signed by Charles Wilks, listing all buildings on the property in 1806. The dimensions are listed as $34' \times 34'$ which are the external measurements of the storage area of the vaulted magazine. It would therefore have been part of the huge building programme undertaken by Wilks between the Board of Ordnance's purchase of Ballincollig in early 1805 and 19th June 1806 (Date of Statement of Property signed by Charles Wilks, WO 55/2285). In the 1828 inventory the vaulted magazine, designated 'OO' and named 'Magazine' is described as being: 'In good internal repair, might be put in order for about £10' (Fig. 3).

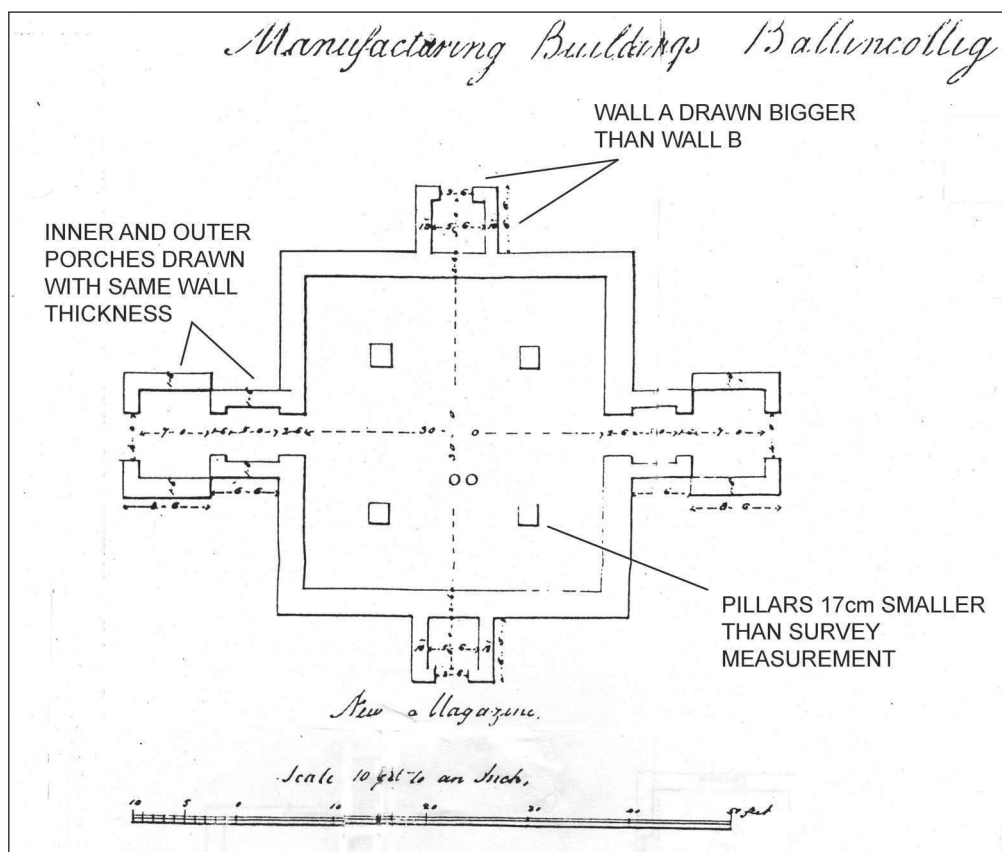


Fig. 3. Drawing of vaulted magazine from 1828 inventory: indicating inaccuracies

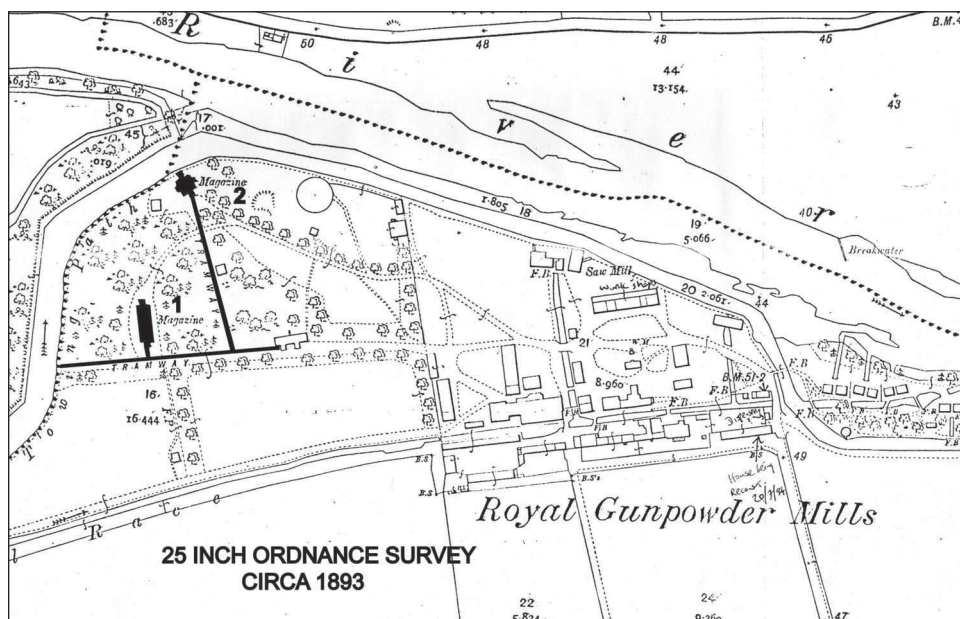


Fig. 4. Showing magazines '1' (most recent) and '2' (vaulted) and tramway between the magazines.
OS sheet number CK 073-11

The most recently built magazine lies 200m south east of the vaulted magazine. It does not appear on the 1828 inventory, nor on the first edition map (1834-1843). Kelleher (1993, 69) states that 'Magazine No 1' (as this third, most recent magazine was designated in correspondence) was built 'after 1834'. He does not suggest how long after. As Tobin showed every sign of being a shrewd business man, he probably used what he had acquired in 1834 and when it was making a profit, started a building programme. Two incorporating mills were built in the 1840s and two in the 1850s (Hamond and Hegarty 1987, 13). The first mention of 'Magazine No 1' is in hindsight when the magazines were granted an exemption from the minimum distance apart set in the Gunpowder Act of 1860 (Kelleher 1993, 69). The long correspondence between the management of the gunpowder mills and Sir Hugh Henry Rose, commander of the British troops in Ireland, on the subject of Fenian threat and threat of volatile materials destroying the Ballincollig barracks is dated 1865-

1870 (Kelleher 1993, 63-71) and refers to 'Magazine No 1'. Thus the most recent magazine was probably built in the 1850s.

SITE LOCATION AND DESCRIPTION OF BUILDING

The vaulted magazine is 300m north west of the primary processing area of the Ballincollig Royal

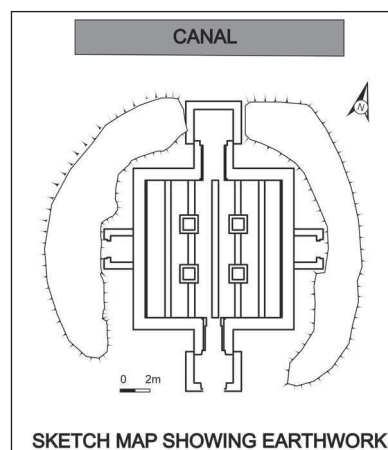


Fig. 5. Adjacent features to magazine

Gunpowder Mills site and 120m north east of the other large magazine. The north of the building is adjacent to the main canal at north and 1m above the water surface at high tide. The building is surrounded by an earthwork (Fig. 5) which was intended to absorb the force of a possible explosion. The following, an extract from the Explosives Act 2005 (based on an act of 1875), states that ‘danger (*sic*) buildings’ should be surrounded by banks of earth:

high as the eaves of the building and 1m wide across the top with the earth sloping at its natural angle of settlement, – about 30 degrees – each side. It is preferable that the earth is free from stones but if it does contain stones these should be no larger than a human fist.

The building consists of a square inner area 9m × 9m, the only openings are the two entrances north and south. The ceiling is groin vaulted, supported on four pillars. To north and south is a barrel vaulted inner porch, with a roofless outer porch beyond this. There are also

sentry boxes on the east and west walls of the building.

The vaulted magazine is constructed out of stone in random rubble with irregularly-sized quoin stones, some quite large. The roof is at present constructed of masonry and mortar although its original form was quite different as discussed below. The smooth rounded lines of

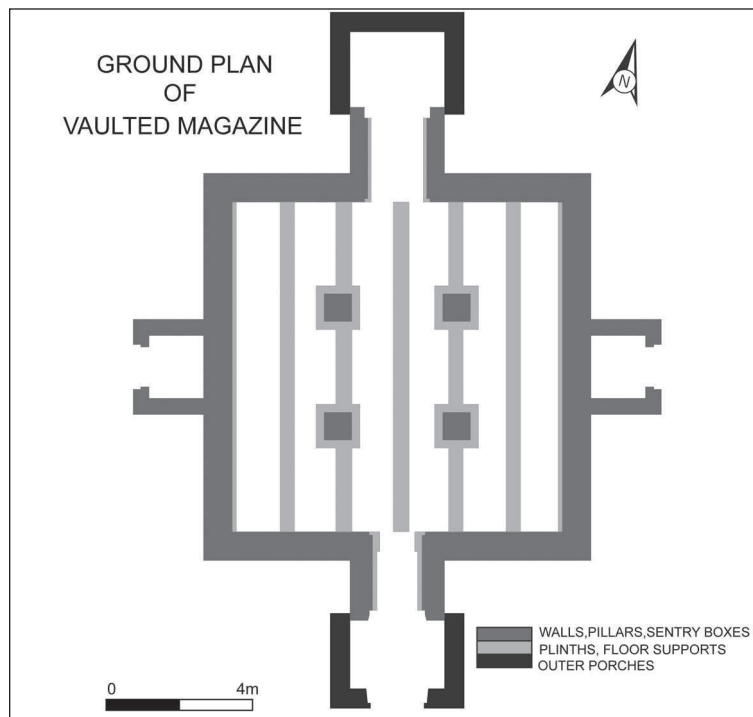


Fig. 6. Ground plan of magazine

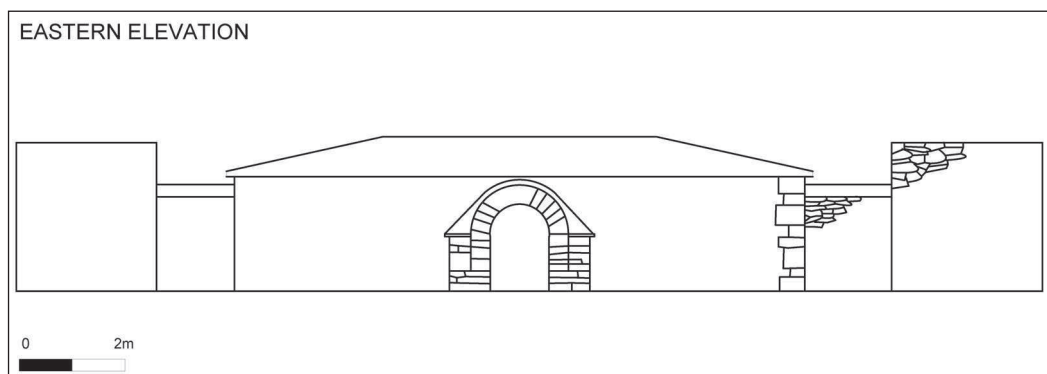


Fig. 7. Eastern elevation: sentry box in centre

the inner vaults are created with a layer of bricks set into the masonry. This brick layer is exposed in the southern porch but is otherwise rendered throughout the building.

The groin vaulting of the magazine springs from the impostes of the pillars. There are three apices in the central vault running between the two entrances at north and south and three

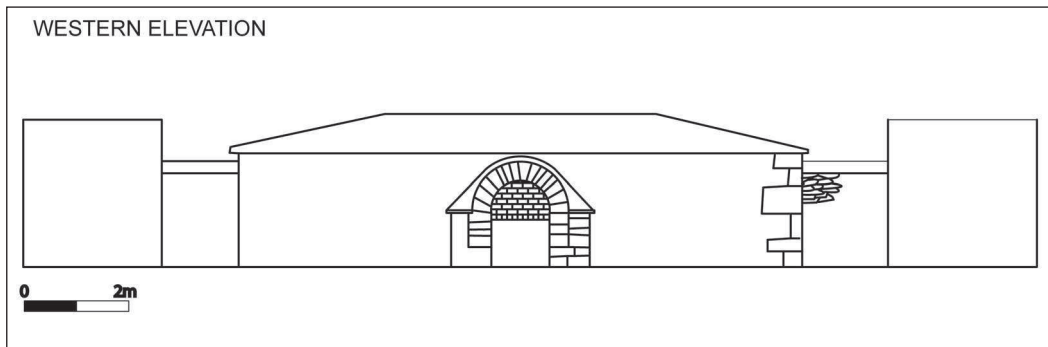


Fig. 8. Western elevation

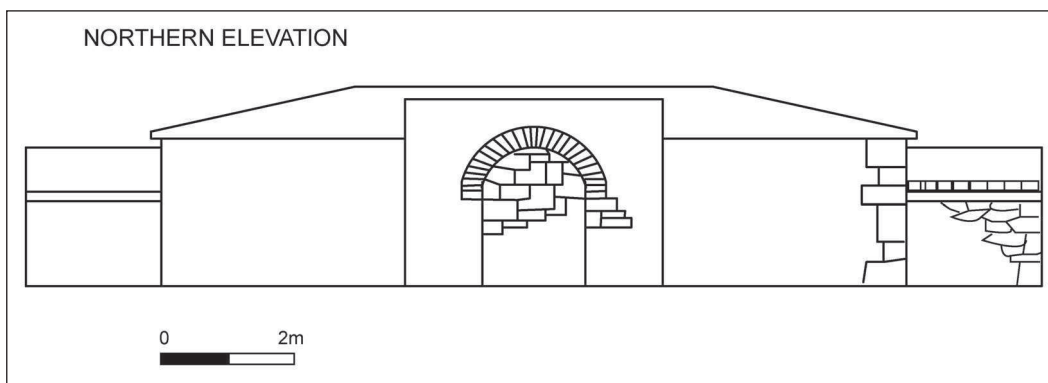


Fig. 9. Northern elevation. Blocked northern entrance faces onto canal

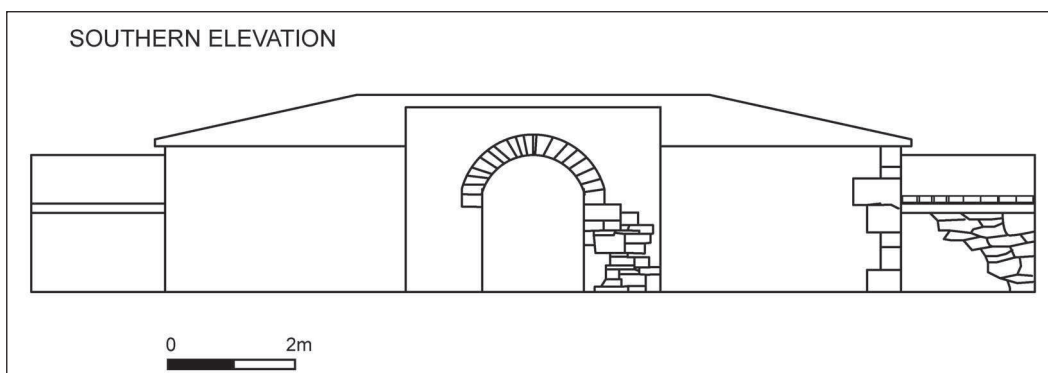


Fig. 10. Southern elevation with entrance porch in centre



Fig. 11. View of interior from north to south, showing pillars meeting at central point (Photo: author)

apexes between the pillars and the east and west walls. The central apexes are formed by four edges fanning out from the impostes.

The lateral apexes are formed by two lines, meeting between the two pillars in the middle and at this meeting point changing from convex edge to a concave corner line.

The pillars are 0.78m square and each stands on a stone plinth 0.30m high. The pillars are rendered over above the imposte but the stones of the pillar below the imposte appear to be painted. There are traces of brown paint on the render while

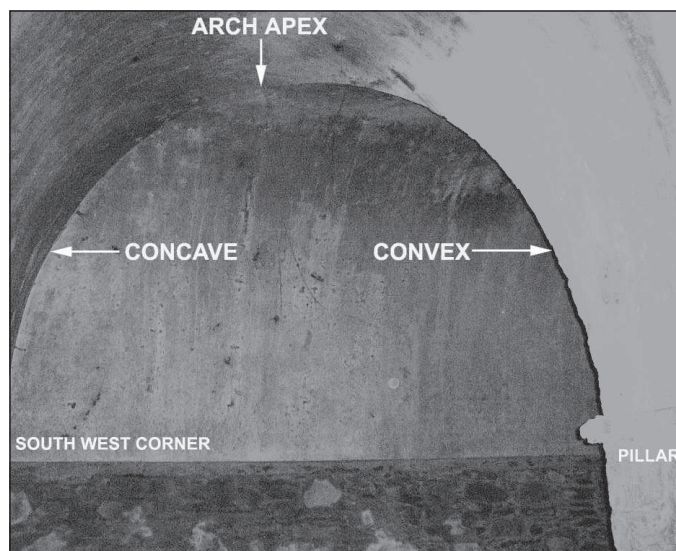


Fig. 12. Line of arch from pillar to south west corner (Photo: author)

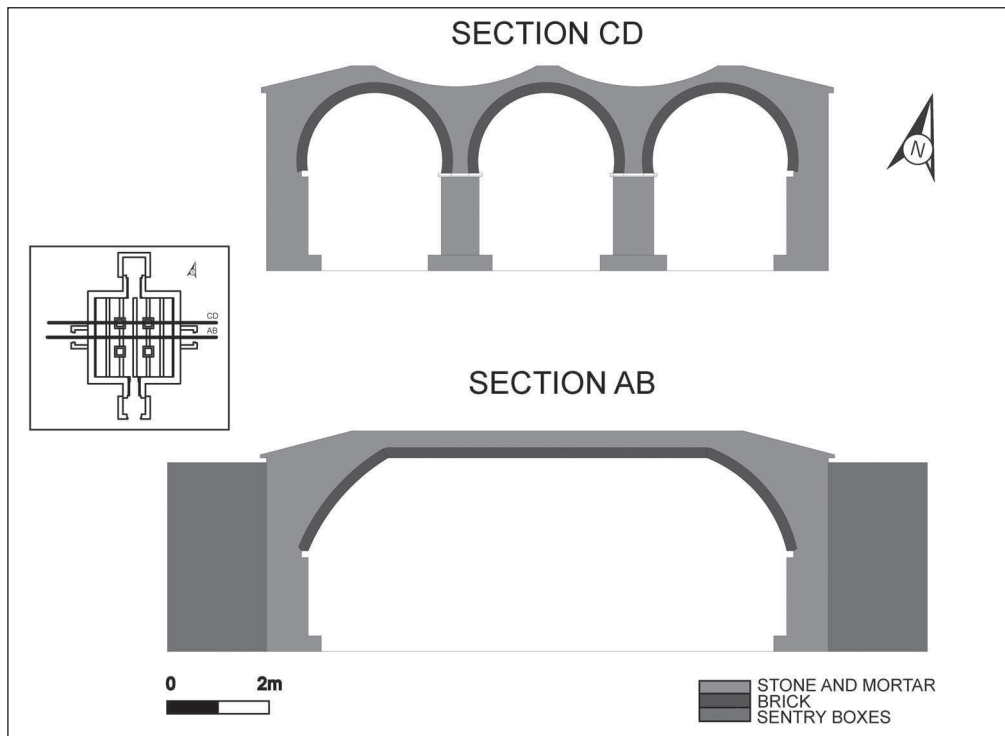


Fig. 13. Sections west to east. CD through pillars, AB through centre of building

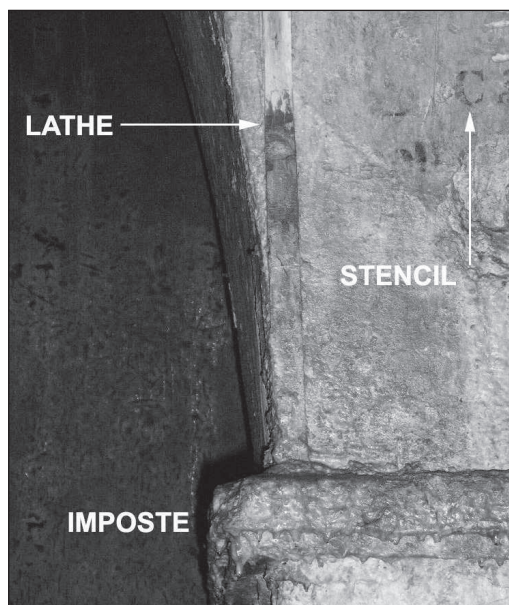


Fig. 14. Features of pillar (Photo: author)

on the south west pillar there are some stencilled letters (Fig. 14).

The interior walls are rendered from the apex of the ceiling to a mid line running around the whole building from the outer entrances of the inner porches. This line is 1.5m above the floor supports. Below the line the walls are of unfaced random rubble. The midline is a trench set into the wall, measuring 12cm top and bottom and 8cm deep. The render overhangs the top of the trench by about 3cm, indicating that some form of wall covering has been removed from the lower section.

On the north western wall, a piece of timber about 1.5m long survives in the trench and has evidence of iron nails in it, suggesting a liner had been attached to the wood which possibly covered the lower part of the wall. Above the midline trench (Fig. 15) there are a considerable number of fixings in the plaster which



Fig. 15. Midline trench (Photo: author)

comprise nails and some wrought iron hooks which have flattened ends. In some places a piece of wooden lathe remains attached by these fixings to the ceiling. These lathes extended to the impostes of the pillars (Fig. 14) and there are fixings at the apex of the vaulting, so whatever was fixed to the plaster appears to have covered the whole ceiling area.

During the 18th and 19th centuries it was common to line Board of Ordnance gunpowder magazines with soft wood matchboarding. At Purfleet, Essex (1760) this was the case (Coad 1989, 248) although copper fixings were used. The 1993 RCHME survey of the 1867 rebuild of the Grand Magazine at Waltham Abbey found evidence in drawings (Fig. 17) and inventories that the building was matchboarded in deal throughout using copper fastenings. In the survey's description of the extant magazine the matchboard lining does not survive, but small timber pieces, projecting from the brick work in the interior remain.

Finally, in the long correspondence between Sir Hugh Henry Rose and the directors of the Royal Gunpowder Mills, Ballincollig, concern-

ing the threat the magazines posed to Ballincollig barracks, there is a reference to the boarding of the magazines. Captain Mackay, Commanding Royal Artillery at Ballincollig, was appointed Inspector of the mills under the Gunpowder Act. In 1867 he reported that the magazines were: 'well and solidly constructed of stone and entirely cased inside with wood' (Kelleher 1993, 64).

The purpose of the wood lining, it is clear, was to minimise sparks from grit but also to protect the gunpowder barrels which were quite easily damaged, being made out of white oak.

Five low stone walls run north to south down the length of the magazine, two of which are interrupted by the pillars. These walls are of random rubble 0.44m wide and approximately 0.20m high, and are punctuated by ventilation holes 18cm square. The ventilation holes are set in two rows running east to west 30cm outside the pillars. Those in the westernmost low wall are out of line but both rows continue through the outer walls where they are now blocked by the present ground level.

Similar low walls also occur in 'Magazine No 1', built by Tobin and in a building about 50m south of the vaulted magazine. This building was indicated in the 1828 survey as 'The Old Press House', in 1876 'Canister Store' and in 1891 as the 'Bagging Shed'. It was also used by Murrays of Cork as a magazine up until the early 1970s. As a canister store and bagging shed it would have needed to be well ventilated and this was the purpose of the low walls: they supported, but allowed airflow, around a wooden floor which can be seen lying transversely across them (Fig. 16).

Powder magazines required good ventilation to keep the powder as dry as possible. The War Office plans of Waltham Abbey, Essex (1867)



Fig. 16. Floor supports in canister store (*Photo: author*)

show the Grand Magazine built on similar stone piers. Purfleet, Essex (1760) was constructed in the same manner. At the Old Arsenal, Baton Rouge, Louisiana, this under floor space was filled with barrels of charcoal to absorb moisture. The wooden floor would have been kept scrupulously free of grit and at Waltham Abbey these oak boards were leather covered (RCHME 1993) to minimise damage to barrels which were moved by rolling and stored on their sides.

The inner porches are barrel vaulted. The north porch is rendered down to the midline but the render has been removed from the southern porch, revealing the underlying structure. The crown of the arch is constructed out of brick, which varies in colour and size and shows a lot of intrusive stone and small shell

which would suggest a local provenance. The lower sections of the arch are constructed of stone, held together with a very thick lime mortar. The midline trench continues from the inner walls of the building to the outside opening of the inner porches. At the main room opening of all four corners of the inner porches a hole 15cm square is cut through the corner, forming a triangle between the two trenches, the inner porch trench and the main room trench. It is possible that this held a piece of wood that reinforced the two pieces of wood in the trenches, in the same fashion as the transverse timbers reinforcing the 'A' of a roof.

Below the midline trench, the wall of the inner porch is constructed with random rubble. On the northern inner porch, the extant plaster hangs out 3cm over the lower wall suggesting



Fig. 19. Southern outer porch (*Photo: author*)

inventory plans, so were clearly built during the Board of Ordnance manufacturing period, 1806-1815.

The outer porches are constructed of random rubble. They are rectangular with the southern and northern entrance walls being 1m wider than the side walls.

The southern, outer porch is entered at the south side via an arched door opening. The arch springs from a rock cut ashlar stone on both sides, and the voussoirs are more even on the outer side. There are traces of a different mortar and a few pieces of brick around the ashlar stones, suggesting that these were inserted after the outer porch was built. On the inside of each ashlar stone is a copper ring, 2cm diameter and 1cm thick. One is a perfect circle: the other is distorted into a spiral.

The door jambs can be seen in outline but there are no traces of door hangers. At Purfleet

magazine in Essex (1760) the doors are oak, but with copper-sheathing secured with copper studs. The magazines at Morice's Yard, Devonport (1740) and Priddy's Hard, Hampshire (1770) also had copper sheathed doors (Coad 1989, 245-262). All of the latter were built by the Board of Ordnance as was the Ballincollig vaulted magazine so it is possible that the Ballincollig example had the same type of doors.

Next to the keystone of the outside of the arch a hook with a flattened end is inserted. It is similar to the hooks in the plastered ceiling of the main room, except it is of copper and the other hooks are wrought iron. It is surprising to find wrought iron in a magazine: copper was generally used for its non-sparking qualities. The inventory of 1828 lists many hundreds of pieces of copper hardware: nails, sheets and on the 1876 map one of the workshops is designated 'Copper store'.



Fig. 20. Western sentry box (Photo: author)

The northern, outer porch would have exited to the mill canal, the bank of which is 2.5m to the north. The arch of this porch is blocked, probably during the 1840s or 1850s when the 'No 1 Magazine' was built. The unstable political situation in Ireland meant that one entrance was easier to secure than two and also at this time, a wooden tramway was built (Fig. 4), connecting the two magazines and the dusting house. The later magazine was not situated directly on the canal but was connected to it by 50m of tramway. The tramway is no longer extant.

There are two sentry boxes, one on the east and one on the west of the building, they each are rectangular in plan and 2.5m and 1.9m in extent. They abut the walls of the main building, and have an arched entrance opening. It is difficult to estimate the original floor level as

the western box is filled with debris but the floor of the eastern box is 1.5m below the present ground surface. The earthwork which surrounds the magazine has partially collapsed against them but when they were in use the doorway would have been about 2m high, so it appears that a guard could stand in the box in the normal fashion.

Both sentry boxes have a midline trench similar to the main building and were probably lined with matchboard. The western box has a brick inset in the top of the arch 0.42m below the keystone which was possibly weather protection against prevailing winds. Perhaps this was to facilitate fixing a door if the sentry box had been utilised as a store at some point. It would be easier to make and insert a rectangular door than a door that would fit into an arched opening.

The roof of the sentry boxes is rendered and was clad in slates, these being set into the render and being narrower than slates found on the roof of the building, some $0.34\text{m} \times 0.52\text{m}$ in extent. There is no sign of fixings for timber battens on the render, however one slate was found with copper nails still in place. At the juncture between the main building and the sentry box roof, a rudimentary flashing of slates is fixed to the main building.

In the western sentry box there are a large number of metal canisters: possibly about 80. Local information has it that these contained powder for sporting guns. This seems possible, as in the last years of manufacturing at the Ballincollig powder mills, one suggestion was to keep the mills going by producing black powder of a grade suitable for sporting guns. Also on the 1876 inspection map the building 50m south (already mentioned) is designated 'Canister Store'. Probably the canisters had some kind of measure at the opening, because there would be too much powder in one canister for one shot.

The top covering of the main magazine building is obviously not the original roof. It is a cross of half rounded masonry dividers, set within a square of similar half rounded masonry edges (Fig. 21). From the central meeting point of the dividers to the outer edge of the square is 3.45m and from the edge of the square to the centre line is 3.45m. From the edges of the outer square, a skirt of original roof remains which is also 3.45m on the corner ridges. In the squares formed by the half rounded masonry dividers, the roof dips down, forming four shallow bowls, the deepest part of which is about 0.50cm lower than the top of the masonry dividers. The masonry comprises broken bricks, small stones and rough mortar. There are a few slates still in position on the roof area, which are a pink/ grey colour, and vary in width from 0.24m to 0.30m. Not all of the slates have nail holes, they are fixed in the render. There is no evidence of nail holes for timber fixings in the masonry.

The architectural explanation for the top covering of the magazine is that the half-rounded masonry dividers are the upper manifestations of the crown apex of the vaulting and the bowls are the centre of the four cornered fan that rises from the impostes of the pillars. The bowls are, thus, necessarily lower.

DISCUSSION

The security of the two magazines was the subject of a long correspondence between Sir Hugh Henry Rose, commander of the British troops in Ireland, and the government in Dublin. The main cause for anxiety was that if the magazines blew up it would cause immense harm to the barracks at Ballincollig. Rose was also concerned at what he perceived to be several security issues: the increase in production at Ballincollig meant a far greater number of barrels stored in the magazines.

According to production figures in Kelleher's history of the gunpowder mills, during the period of the correspondence 30,000 barrels were produced a year, that is, a weekly production of 600 barrels. In a public meeting in 1867 complaints were made that 5,000 barrels were being held in the magazines.

Indeed, the magazines could hold 5,000 barrels but whether this was a constant volume is questionable. The inner space of the vaulted magazine, the subject of this survey, is nine metres square. The drawing of the Waltham Abbey Grand Magazine, which is to scale, shows four barrels lying on their sides in a space of six feet. This would mean each barrel was eighteen inches (or approximately 0.50m) in length. Replica barrels used for demonstration purposes at Royal Armoury museums are 24" (Alan Gosling, Curator, Purfleet, Essex, pers. comm.) Gunpowder is a very dense, heavy substance therefore the normal storage weight of 100lbs could be contained in a much smaller barrel than a full-sized beer barrel. Thus if a gunpowder barrel on its side measured 0.50m long and possibly 0.35m at



Fig. 21. Masonry dividers on roof. (Photo: author)

the widest part, then a maximum of 360 would fit on the first tier. This allows for a central access passage of 1m width, the bottom area of the pillar plinths and five wooden partitions for air circulation. Above the 2m high midline the vaults would impede storage so perhaps up to six tiers would be involved. This means 2,160 barrels could be accommodated. The more recent magazine, being rectangular, and about double the size, could therefore hold about

4,000 barrels. However, Mackey, the government-appointed inspector, in 1868, found one magazine to be empty and the other holding 800 barrels (Kelleher 1993, 70). So therefore, although maybe every so often the magazines reached the full capacity of 5,000-6,000 barrels (approximately ten weeks production) waiting for shipment, if the annual production was only 30,000 barrels, there must have been periods of much lower capacity.

Rose's second fear was that of attack or theft of gunpowder by nationalist groups. His concern was always to move the troops from Ballincollig. He also felt the newer magazine was too lightly constructed. Finally, in 1870, the War Office, who had never wanted to move troops out of Ballincollig, sent a team of inspectors to the powdermills. They recommended the removal of the solid masonry roof of the vaulted magazine and its replacement by a less substantial structure. Soon after the masonry roof was removed and replaced by a timber and slate roof (Kelleher 1993, 70).

The second roof was dismantled in 1946, probably for secondary re-use. The exact structure of both the 1806 and the 1870 roofs is therefore a subject for conjecture. According to Rynne (pers. comm.) the charge house roof is stone, set in mortar and built up in a corbelled construction. This would be very appropriate for a round building and would be a use of tra-

ditional Irish stone masonry techniques. The Board of Ordnance magazines of the 18th century used an all brick construction. Priddy's Hard has brick-built barrel vaults and has a traditional gable roof of slate (Coad 1989, 262). Unfortunately Coad does not give details of the construction of the roof. At Purfleet the barrel vaults are constructed of brick which is brought to a horizontal line above the vaults. On top of this lie 12 tons of silver sand and finally a conventionally constructed roof (Fig. 22).

The 1885 plan of Waltham Abbey Grand Magazine shows a construction with a barrel vault three bricks deep and above this Portland Cement in a 3:1 mixture. It also shows the 'tiles' to be set in cement similar to the extant roof at Ballincollig. It is quite likely that the original roof of the Ballincollig vaulted magazine was of stone strengthened by thick mortar and then rendered, with the slates set into this. Brick is sparingly used in the building,

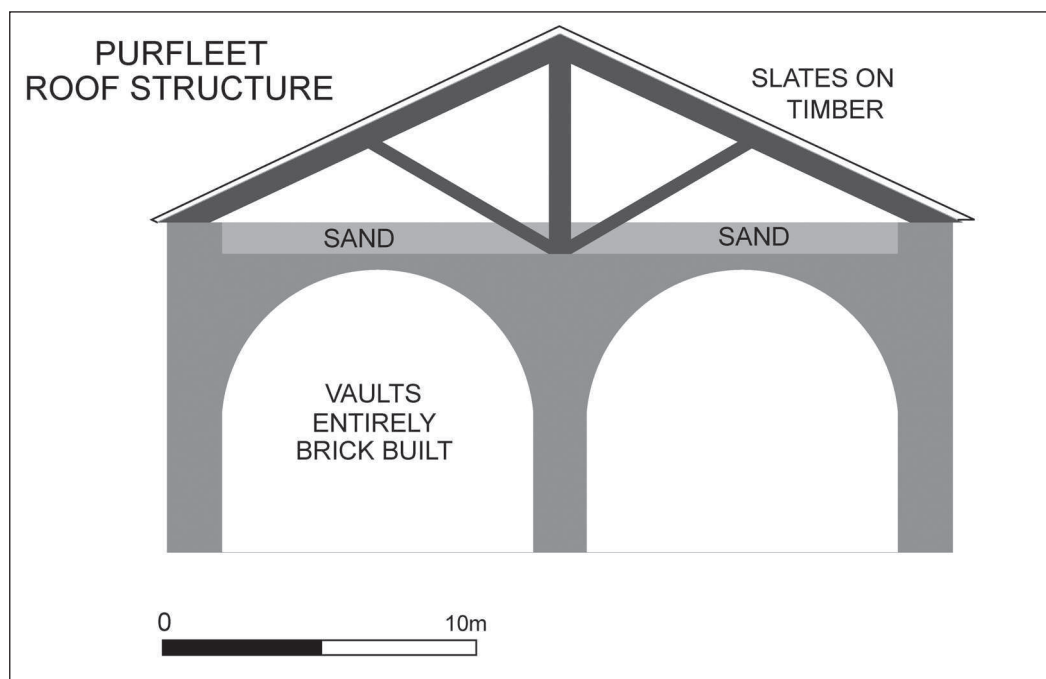


Fig. 22. Roof at Purfleet (only baseline to scale): reconstruction based on personal comments by Alan Gosling, Chief Curator, Purfleet Magazine Heritage Centre

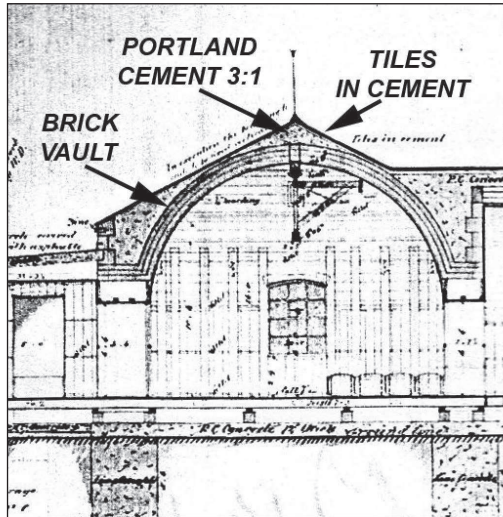


Fig. 23. Roof structure at Waltham Abbey
Grand Magazine

in the barrel-vaulted inner porch it is only used for the top of the barrel vault, the rest is of stone. Brick is probably used in the same manner in the vaulting, to achieve geometric precision. It would be a skin with stone and mortar as fill on top (as at Waltham Abbey) (Fig. 23).

The 'skirt' of mortar around the existing roof covering is no doubt part of Wilks' masonry roof, it could not be removed in 1870 because it formed part of the whole structure of the building. There is very little sign of disturbance on the extant roof. This lack of disturbance might indicate that a cushioning substance lay between the extant roof and the masonry roof which was removed. Is it possible that there was a sand filler between the two? If so, this would have also levelled the dips formed by the pillar tops (Fig. 25).

The fixings of the 1870 slate and timber roof

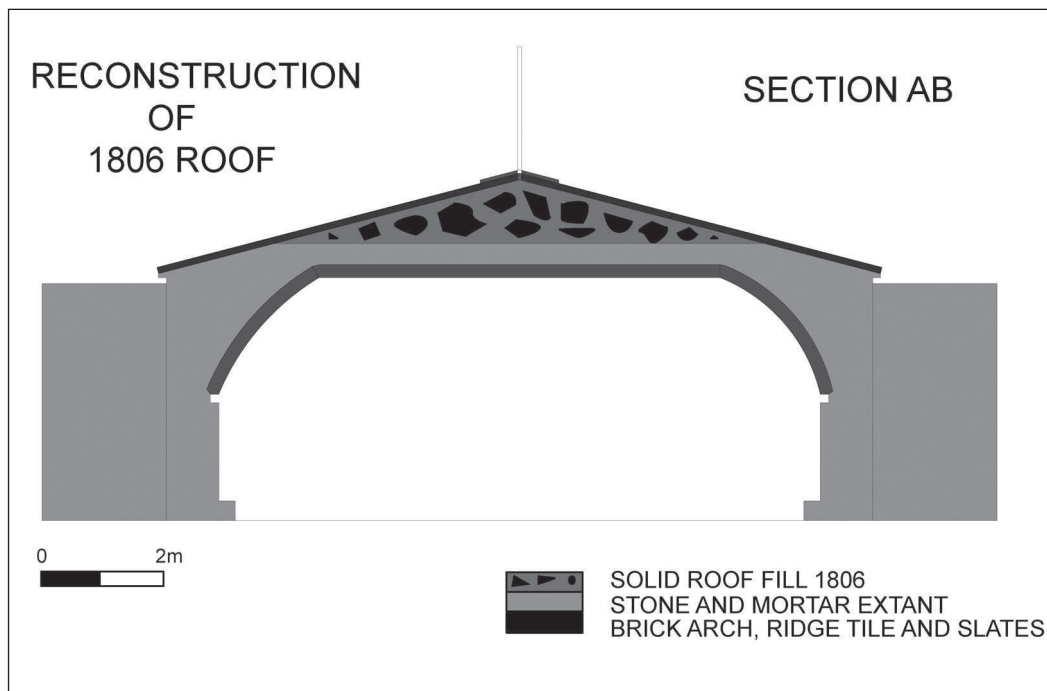


Fig. 24. Probable roof construction of Ballincollig magazine central section
including lightning conductor

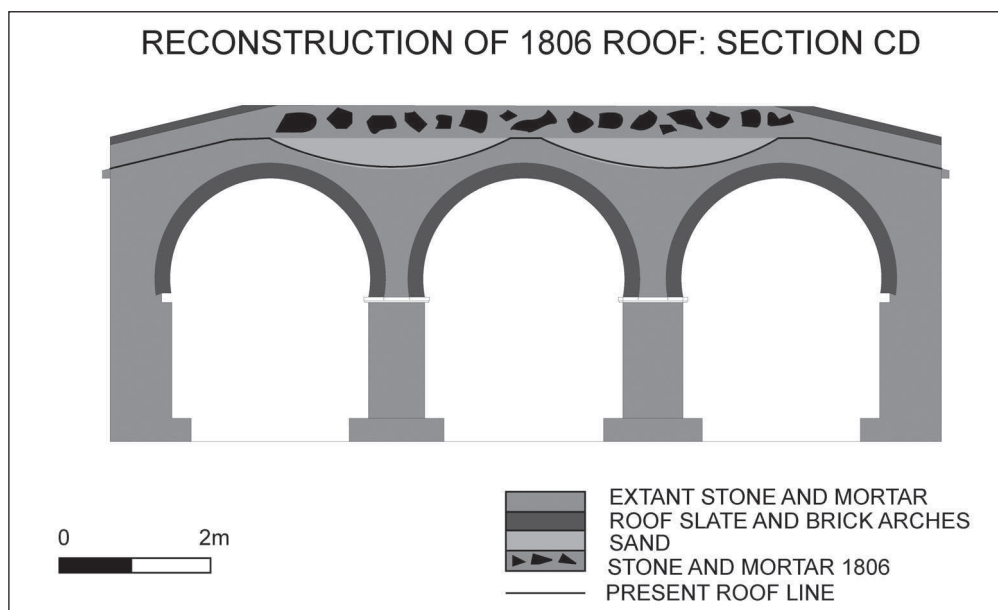


Fig. 25. Probable 1806 roof construction of section through pillars of magazine

are a subject for conjecture. The only possibility would be to fix a timber all around the square, have a central post and two diagonal 'A' frames running NW/SE and NE/SW, which would form the ridges. The fact that the 1870 slates were removed in 1947, but not the slates on the remaining masonry, supports the theory of the lower slates being original, as they were rendered they would have needed cleaning and would have been more difficult to remove.

Fifty years separate the two extant magazines at Ballincollig. The difference in their structure mirrors the difference in the political situation and the difference in the productive purpose of the gunpowder mills. The vaulted magazine was built when attack from the outside was a probability, the strength of its structure was intended to withstand bombardment. The purpose of the magazine built, possibly in the 1840s, as part of a commercial enterprise in peacetime, was to deter nationalist factions within the locality from pilfering small amounts of gun powder. Such factions were un-

likely to mount an artillery attack.

Coad points out that magazines built after the Napoleonic wars were 'not expected to withstand bombardment and were built without the massive expensive brick vaults' (1989, 263). Although of course, as can be seen in the Table 1 below, vaulting was still a common building form but with a constructional skin of brick and above it a concrete masonry roof. Rose, the commander of the British forces in Ireland, complained of the lightness of the structure of the later magazine but this lightness was probably based on the same principle as building the incorporating mills out of wood i.e. less damage on explosion. Also commercial enterprises, unlike government ventures, have to make a profit and a lighter structure would have been cheaper to build.

The development of the great Ordnance dockyards in the 18th century at Devonport, Portsmouth and Chatham made naval architecture a subject of widespread interest. Discussing Morice yard, Devonport, Coad refers to

a tradition that Vanbrugh was the architect (1989, 252). Although documentary evidence does not bear this out, it demonstrates that military architecture was seen as well designed and stylish rather than basic and utilitarian as are the army camps of today.

An article in *The Gentleman's Magazine* 1760 discusses the efficacy of different types of vault to absorb shelling from enemy fire. It must be remembered that up to 1815, this was a real possibility. It would have been a particular consideration of Charles Wilks, in the year of the Battle of Trafalgar, and with the rising of 1798 only seven years before.

OF POWDER MAGAZINES By M. Belidor from *Gentlemen's Magazine*, Vol. XXX, 1760, pp. 264, 269-70.

Of all military edifices, there are none of greater consequence, or that require more judgment to be properly constructed than powder magazines: For as they are, or should be always vaulted, first we should know what kind of arch is most capable of resisting the force of bombs falling on them, whether the semi-circular, the segment, or ellipsis, or the arch of three points: and, secondly, find the proper thickness of the piers to support and resist the trust of the arch. Engineers are mostly divided on the manner of vaulting powder magazines; some pretend that the semi-circular is the best of all, and others say the arch of three points is preferable. Certain it is that the latter has less thrust than a semi-circular and the semi-circular less than the segment, which may be geometrically demonstrated, and I shall, in as plain a manner as possible, prove that the arch of three points has less thrust than the semi-circular, see illustrations at opening of article which show comparative sections of powder magazines (below). Another common defect of semi-circular arched magazines is, that the roof is too flat, by which they receive more damage from the fall of shells, than if they were more inclined; & if to avoid this defeat you make the roof more acute, the key of the arch is loaded with more masonry than needful, and the pins must be thicker to support that weight.

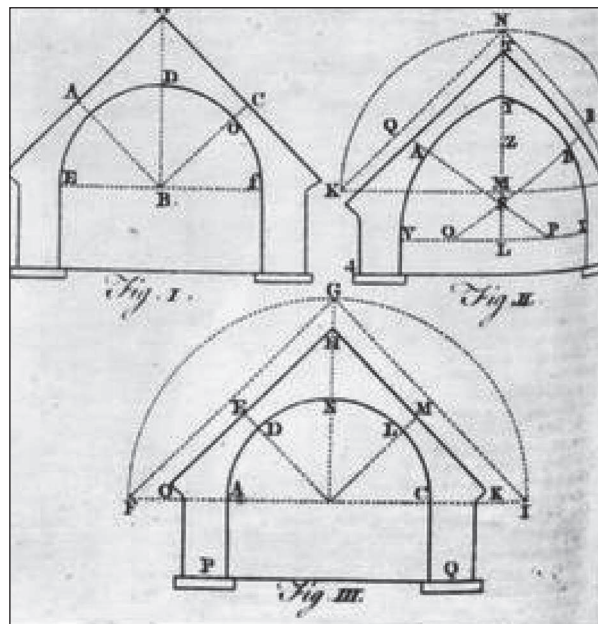


Fig. 26. Illustration accompanying Belidor article. Barrel and groin vaulted magazines

The majority of magazines were built with semi-circular barrel vaults rather than the three-pointed groin vaults recommended by Belidor.

Doubtless this was because less technical expertise was required to build a barrel vault than a groin vault. This is a small sample but, unfortunately, at many large gunpowder mills or ordnance yards the magazines are no longer extant, as at Faversham, Low Wood and New Sedgewick in Cumbria, Chilworth, Surrey (only a very small expense magazine) and even at the huge Du Pont complex at Hagley, Delaware.

Thus it appears that when Charles Wilks designed and built the magazine at Ballincollig he produced a unique, although recommended, example of the genre. The aesthetic effect is enhanced by the building being square rather than rectangular: all the magazines in the table above are rectangular. The suggested reasons for his choice of such an unusual form of magazine are firstly that as an experienced and able clerk of works at three Board of Ordnance sites in

TABLE 1

DATE	NAME	VAULT
1681	Charles Fort, Kinsale, Co. Cork	Barrel: semi-circular
1714	Williamsburg, Virginia	Barrel: semi-circular
1740	Morice Yard, Devonport, Devon	Barrel: semi-circular
1760	Purfleet, Essex	Barrel: semi-circular
1770	Priddy's Hard, Gosport, Hampshire	Barrel: semi-circular
1814	Marchwood, Southampton, Hampshire	Barrel: semi-circular
1860	Beechworth, Australia	Barrel: semi-circular
1867	Waltham Abbey, Essex	Barrel: semi-circular
1830	Baton Rouge, Louisiana	Groin: three pointed
1860	Crown Hill, Plymouth, Devon	Groin: three pointed

England he must have been aware of the three point and barrel vault debate. In wartime and in an unstable local situation, he decided to choose the design best able to absorb shelling. Secondly, as a newly appointed superintendent in what to him must have seemed a very foreign country, he had the confidence to experiment. Often expatriation can liberate creativity because the expatriate is no longer bound by the *mores* of his home country. Thirdly, it was likely that another wider controversy influenced his choice of construction. The 18th century controversy between Romantic and Classical manifesting itself architecturally in the Gothic Revival is encapsulated in the vaulted magazine. It is a Gothic crypt with pointed arches and yet it is as perfectly proportioned as the Parthenon.

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